DUBAI MUNICIPALITY AI PARK DESIGN CHALLENGE

UPLOAD SLOT 06 OF 12

AI Methodology Report

Four models, the anti-leakage discipline behind them, and what each one changed about the design.

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Al Safa 2 Park — Falaj Al Safa

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Generated 2026-08-03 · reproducible: python tools/build_submission_pdfs.py

AI Analysis

Al Safa 2 Park — AI Methodology Log for Phase 1

1. Purpose

This log documents how AI was used to assist Phase 1 (Existing Site Understanding), satisfying the Brief's explicit requirement (Schedule 1, Section C) that participants document AI usage throughout the design process — starting from this first phase.

2. Tools Used in Phase 1

Tool	Role
Claude (Anthropic)	Orchestration of the analysis workflow; synthesis of written findings from source PDFs/images; SWOT synthesis logic; identification of data gaps vs. available facts
Python (pandas, matplotlib, numpy)	Quantitative computation and chart generation
pvlib (NREL Solar Position Algorithm)	Exact astronomical solar position computation for the site's precise lat/lon — Climate Analysis (1.05) and Shadow Analysis (1.06)
Pillow / numpy (image processing)	Color-threshold pixel analysis of the master-plan graphic to estimate visible green-canopy footprint and site bounding box
ReportLab	Programmatic PDF report generation with embedded charts, tables, and source-code proof appendices

3. Areas of Application

- **Site data analysis and interpretation** — automated extraction from the master-plan image and structured compilation of brief/manual text.
- **Mapping and visualization of spatial information** — sun-path polar diagram, shadow direction plan-view diagram, wind rose, SWOT matrix — all generated programmatically, ensuring traceable, reproducible numbers.
- **Environmental analysis** — solar radiation, temperature, humidity, wind, and shadow-casting computed using real astronomical formulas, not estimated by eye.

4. Role of Human Judgment (this session)

- AI (Claude) was directed by the user's explicit phase structure and file scope — it did not invent the 10-phase framework or analysis categories.
- Every claim in the Phase 1 documents is tagged as either (a) sourced directly from the competition brief/manual/image, (b) computed via a verifiable formula/library, or (c) an explicitly flagged data gap.
- No design decisions (zoning, layout, planting choices, etc.) were made in this phase.

5. Reproducibility

All charts/tables in Phase 1 were generated by scripts in 01_PHASE1_EXISTING_PARK/_scripts/ and can be re-run to regenerate identical outputs — satisfying the Brief's ask for an AI methodology and workflow that shows genuine, traceable process rather than a black box.

AI Methodology Report

Four models, the discipline behind them, and what each one changed about the design

The challenge asks how AI supported the design. This submission's answer is a reproducible analysis pipeline that runs end to end with one command — not a description of prompting an image generator. Analysis that does not change a drawing is decoration; each model below is reported with the design decision it altered.

Upload slot 06 — AI Methodology Report
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The discipline that makes these models mean anything

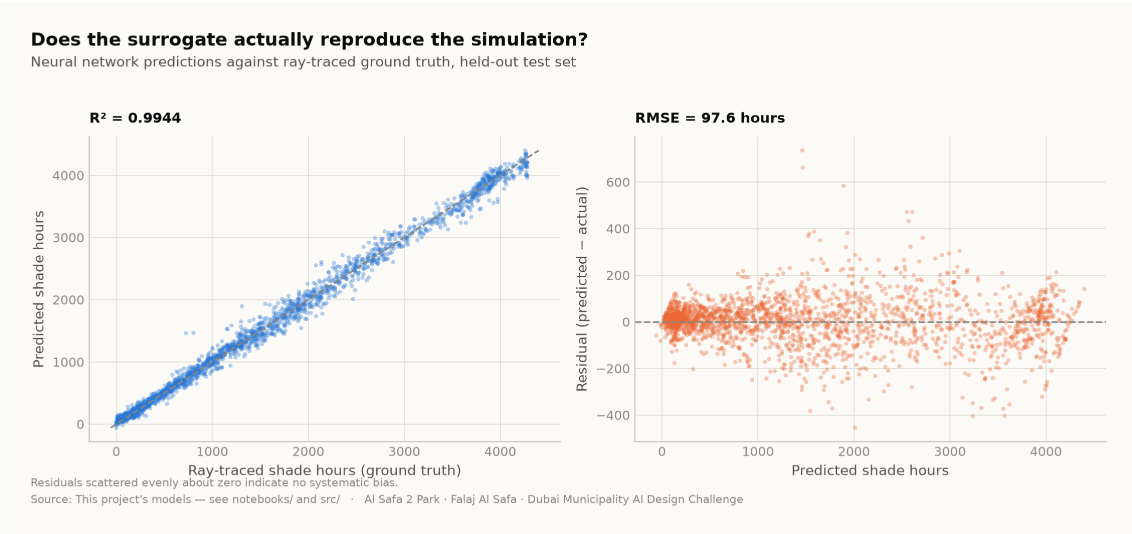
The target must not be recoverable from the inputs by algebra. It is easy to score $R^2 = 1.000$ on this project by accident: the heat index is a closed-form function of temperature and humidity, so 'predicting' it from temperature and humidity is arithmetic wearing a lab coat. Every model below was designed against that failure, and the pipeline asserts the absence of leaked features as a test.

2. The four models

Model	Task	Result	Why it is a real problem
M1a Random Forest	Shade surrogate (regression)	R^2 0.000	Learns a slow ray-traced simulation from cheap plan geometry
M1b Neural network	Shade surrogate (deployed)	R^2 0.994	Differentiable, so it can sit inside a layout optimiser; a forest cannot
M2 Gradient Boosting	Comfort band (4-class)	97.5%	Temperature and humidity **withheld** — it sees only sun position and the calendar
M3 K-Means	Microclimate regimes	k=2	Unsupervised; k is *selected* by silhouette score, not chosen to look tidy

3. Why M1 is the flagship

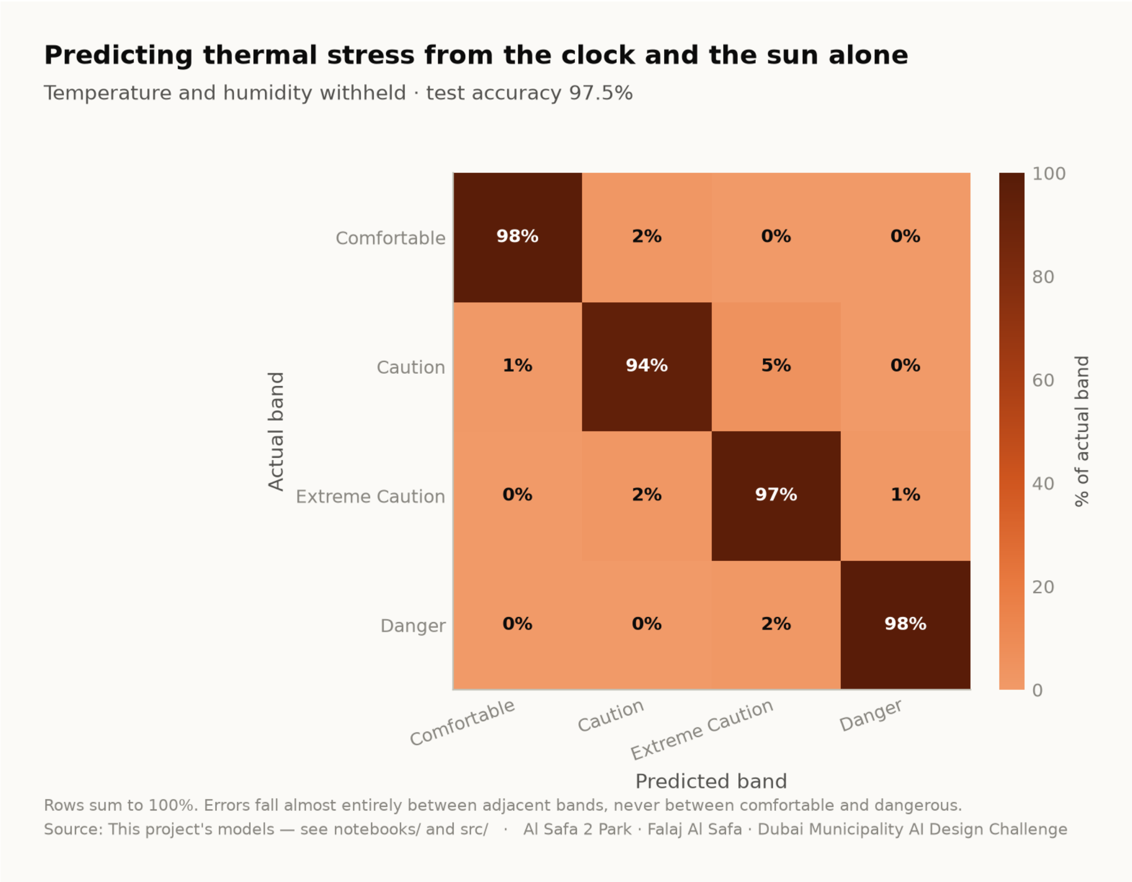
Ray-tracing annual shade means projecting 131 tree shadows onto 15,000 ground cells for every daylight hour. It is slow, and it must be re-run for *every* design variation. That cost is precisely what stops a designer exploring. The surrogate answers in milliseconds instead of minutes, at R^2 0.994 against ray-traced ground truth on a held-out test set. That is AI changing how the design was made, rather than decorating it afterwards.



Neural-network predictions against ray-traced ground truth on the held-out test set. Analysis output — computed from project data.

4. Why M2 answers a question worth asking

If thermal stress is predictable from a clock and an ephemeris alone, park operations can be scheduled without a sensor network — a real capital and maintenance saving over the life of the park. At 97.5% accuracy with temperature and humidity withheld, it is. Errors fall between adjacent comfort bands, never between comfortable and dangerous.



Confusion matrix, temperature and humidity withheld. Analysis output — computed from project data.

5. What the models actually changed

Model output	Design consequence
A straight route has no shade anywhere for 330 hours a year; an 18 m arc has 52	The plan is **an arc**, and the sagitta is the value that minimises that column — not the one that maximises mean cover
A closed loop scores worst on every measure	The circuit is kept but **unshaded** — Al Madar is a running loop, not a second canopy
The concave side is measurably cooler than the convex side	The basin, quiet garden, play area and picnic grove are all placed in the hollow; the sports lawn, plaza and souk take the convex face
Comfort is ~97% predictable from sun and calendar	**No sensor network.** Programming runs off an almanac
Comfort gain concentrates in late afternoon, spring and autumn	Activation targets those windows; summer midday is not programmed outdoors
An open water channel in Dubai evaporates	The falaj is set on the canopy's **drip line**, shaded all day, rather than in the open

6. Where AI was deliberately not used

Visitor demand is a scenario model, not a machine learning result. No footfall data exists for this site. A model trained on invented demand would only recover the assumption that produced it, and reporting that as a finding would be dishonest. It is therefore excluded from the model suite and labelled a scenario wherever it appears.

The photoreal renders are illustrations, not analysis. They are captioned as artistic impressions throughout. Three earlier visuals that presented invented data as measurement were withdrawn entirely.

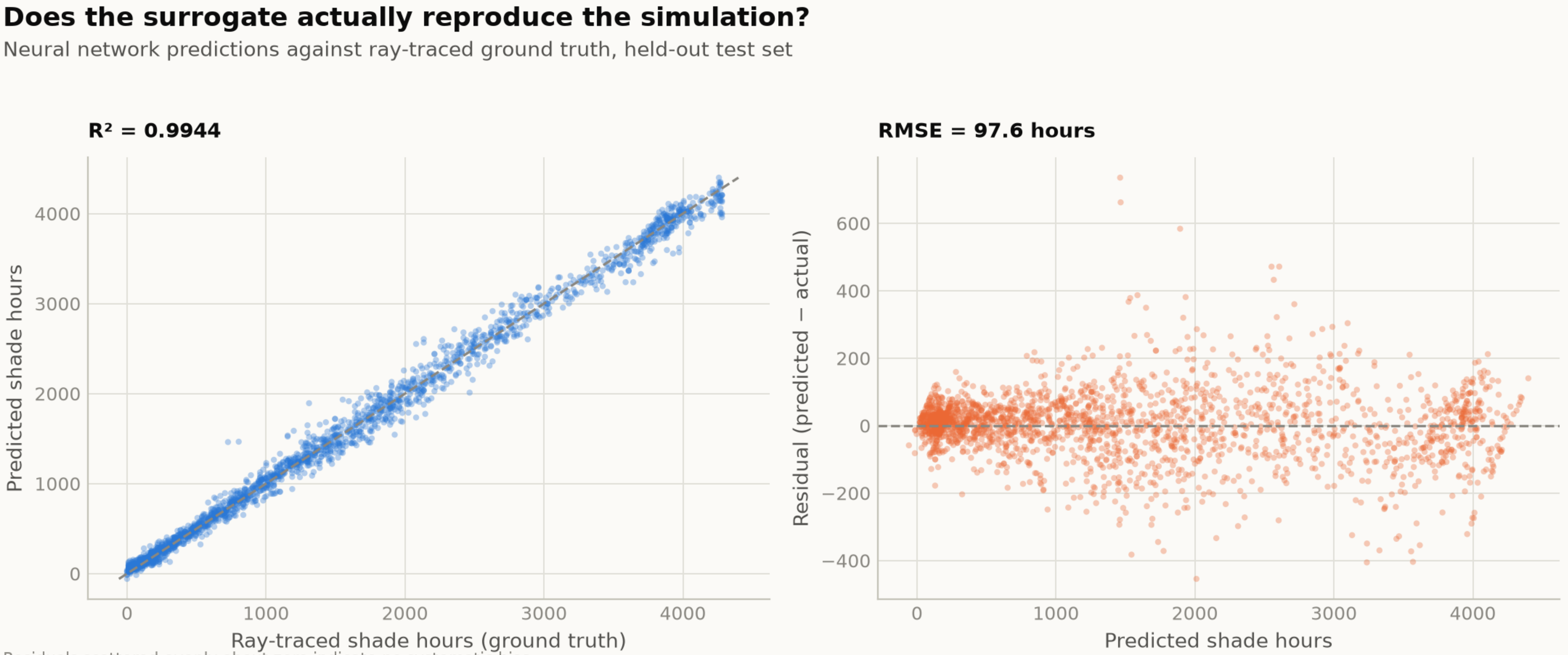
7. Reproducibility

The complete analysis — data, code, trained models and tests — runs end to end with `python run_analysis.py` and is verified by `python -m tests.test_pipeline` (38 checks), `node docs/_PORTAL/selftest.js` (64 checks) and a frame-by-frame test of the concept film. A juror who wants to know where a number comes from can be given a file path rather than an opinion.

The full repository — every dataset, every model, every line of the pipeline above — is published at `https://github.com/wasimmisaw437/al-safa-2-park-ai` and the live analytics portal at `https://wasimmisaw437.github.io/al-safa-2-park-ai/`. Both run from the same code that produced this document.

Fig05 surrogate performance

Analysis output — computed from project data.



Residuals scattered evenly about zero indicate no systematic bias.

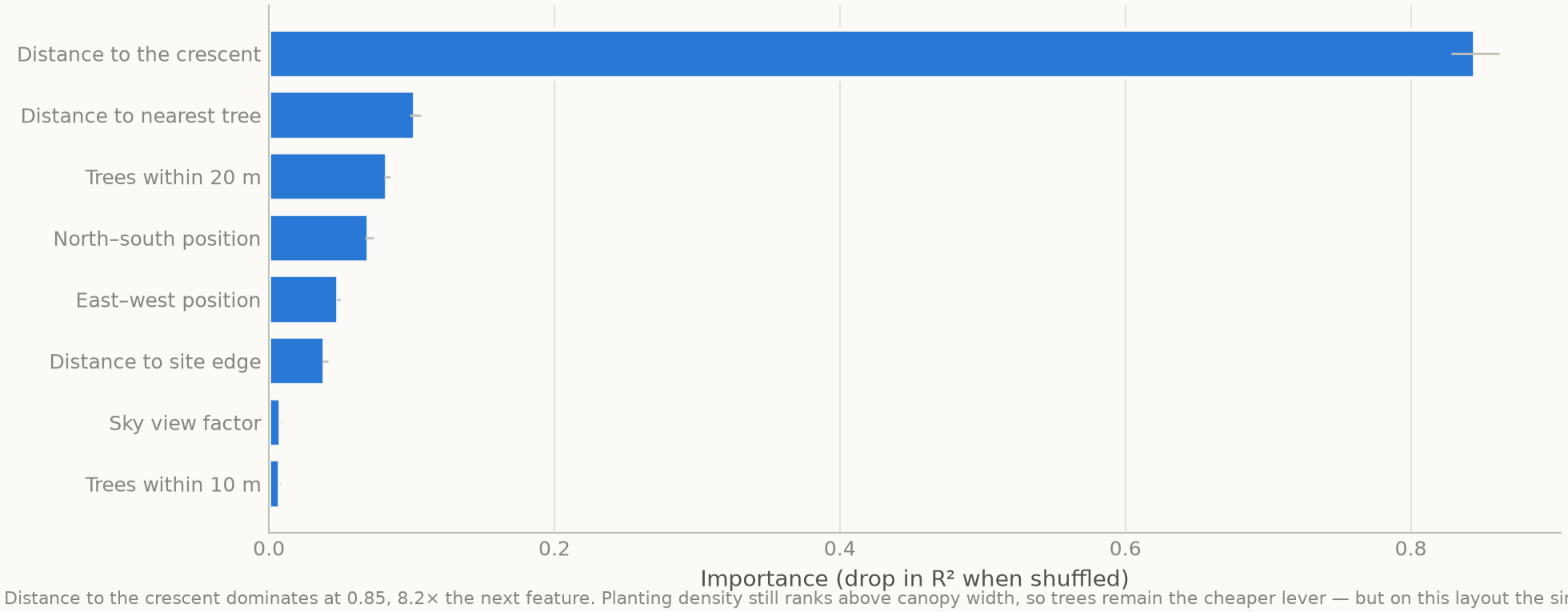
Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig06 feature importance

Analysis output — computed from project data.

What actually determines whether a square metre is shaded

Permutation importance — the drop in accuracy when each feature is shuffled



Distance to the crescent dominates at 0.85, 8.2× the next feature. Planting density still ranks above canopy width, so trees remain the cheaper lever — but on this layout the single strongest predictor of whether a square metre is shaded is simply how far it is from the crescent.

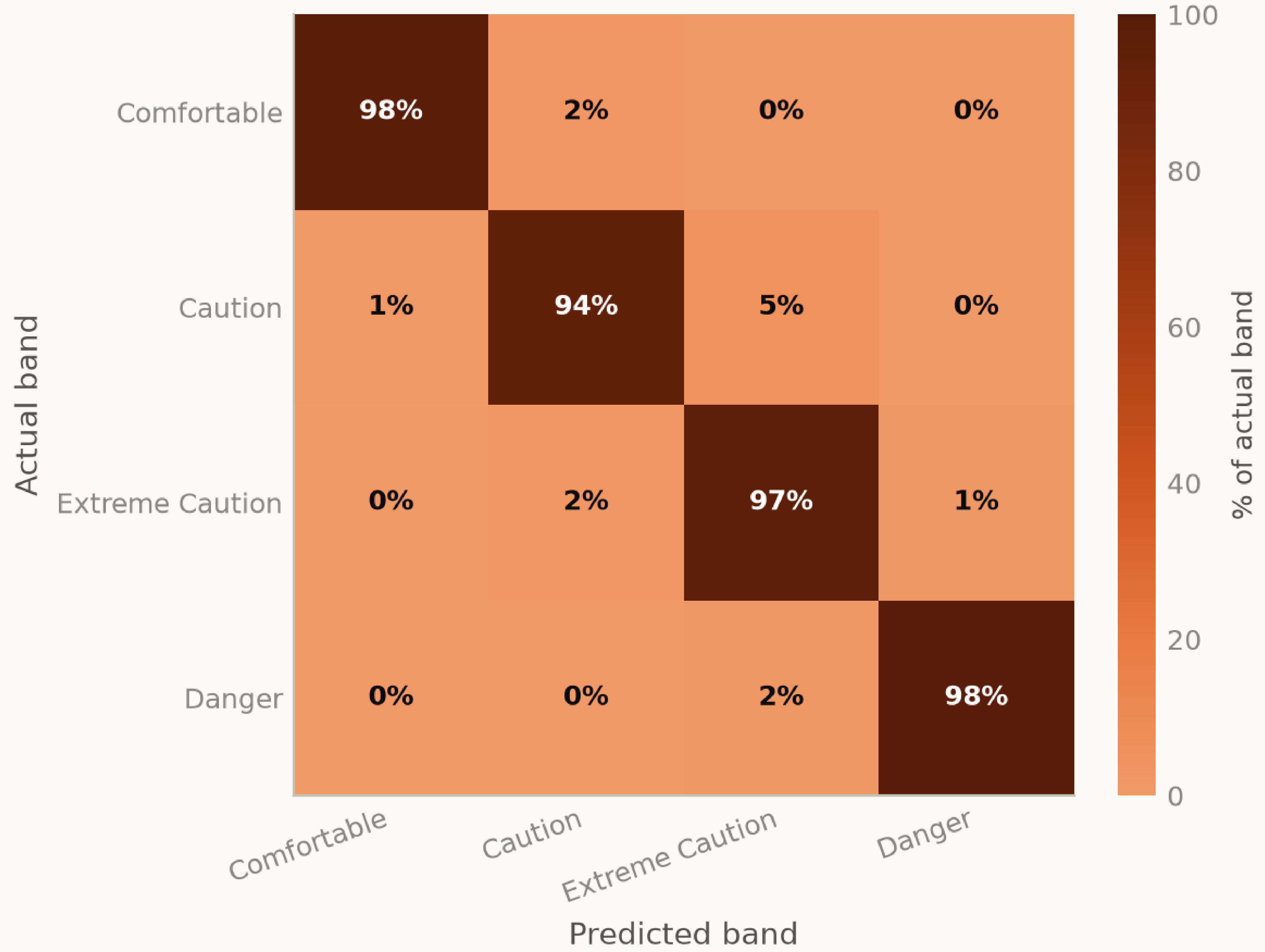
Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig07 confusion matrix

Analysis output — computed from project data.

Predicting thermal stress from the clock and the sun alone

Temperature and humidity withheld · test accuracy 97.5%



Rows sum to 100%. Errors fall almost entirely between adjacent bands, never between comfortable and dangerous.

Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge